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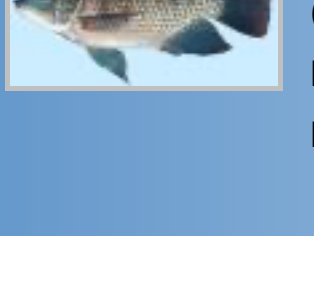
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Oreochromis aureus

(Blue Tilapia)

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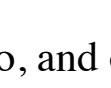


Howard Jelks - U.S. Geological Survey

Oreochromis aureus (Steindachner, 1864)

Common name: Blue Tilapia

Synonyms and Other Names: *Sarotherodon aurea*, *Tilapia aurea*; Israeli tilapia

Taxonomy: available through 

Identification: In general, cichlids (Cichlidae) are superficially similar to sunfishes and black basses (*Lepomis* and *Micropterus*; family Centrarchidae). Cichlids can be distinguished from centrarchids by a single nostril opening on each side of the head (vs. two in centrarchids) and the presence of a discontinuous or two-part lateral line (vs. continuous in centrarchids).

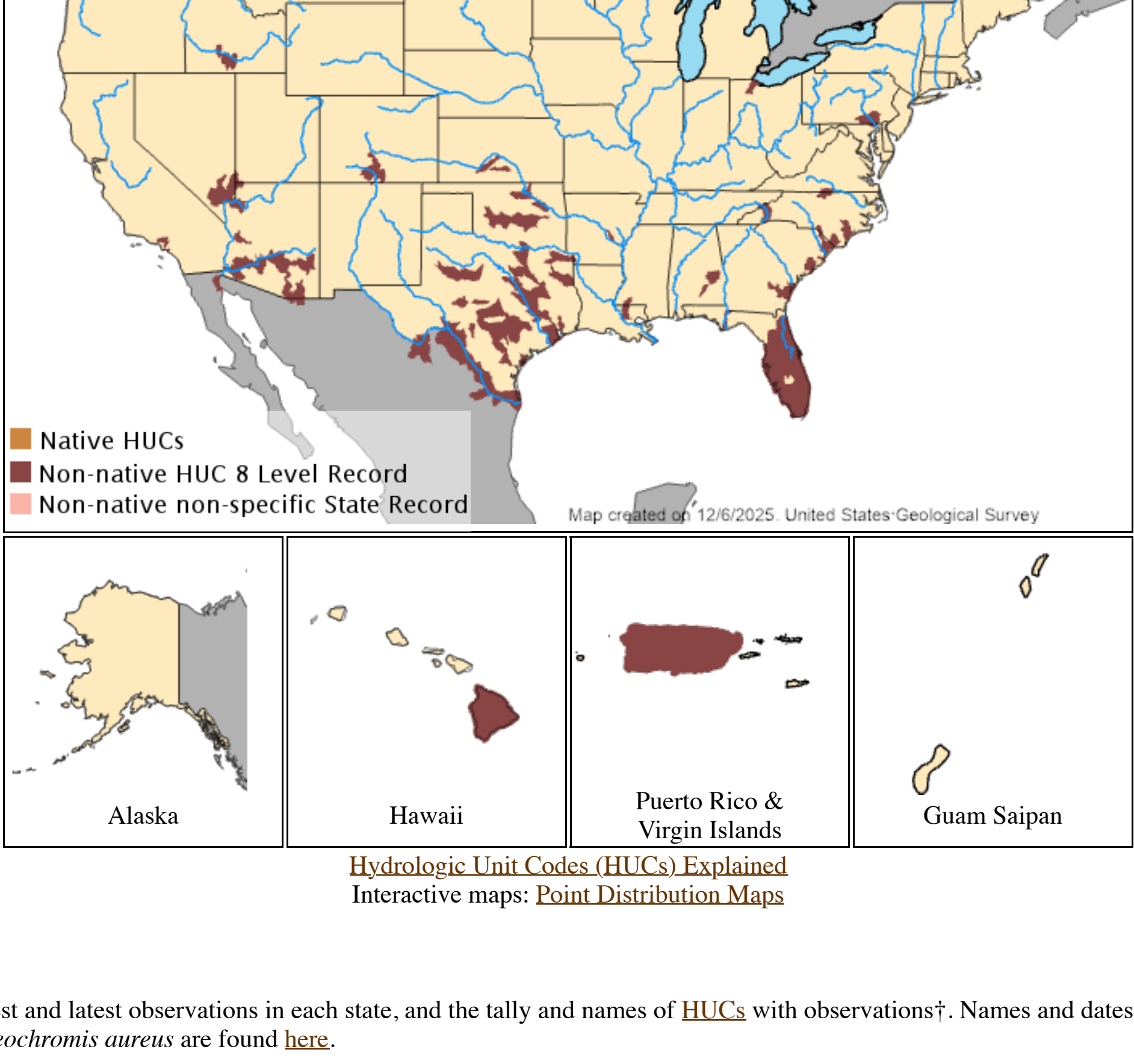
Distinguishing characteristics, synonyms, photographs, keys, and discussion of hybrids were provided in Trewavas (1983); for identification also see Page and Burr (1991), and Skelton (1993). Illustrations and diagnoses of larval and small juveniles of introduced populations were given by McGowan (1988). Color photographs were presented in Axelrod et al. (1985) and Axelrod (1993). Many or most accounts of "*Tilapia nilotica*" in U.S. ponds probably refer to *O. aureus*, likely imported from Israel, before the two species were shown to be distinct (Trewavas 1983).

Blue Tilapia is closely related to, and often confused with, Nile Tilapia (*O. niloticus*). These two species can generally be distinguished by the following characteristics (Trewavas 1983):

	Blue Tilapia	Nile Tilapia
Number of dorsal spines	15-16 (mode 16)	16-18 (mode 17)
Total dorsal fin rays	27-30 (mode 28)	29-31 (mode 30)
Banding on caudal fin	No distinct bands/stripes	Distinct, regular dark stripes
Male breeding coloration	Metallic blue	Red
Dorsal fin margin	Vermillion	Dark grey or black

Size: 51 cm (Lee et al. 1980 et seq.

Native Range: Tropical and subtropical Africa, and Middle East. Native range includes Senegal, Niger, and many smaller drainages and lakes in Africa and Middle East (Trewavas 1983; Skelton 1993).



Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Oreochromis aureus* are found [here](#).

State	First Observed	Last Observed	Total HUCs with observations?	HUCs with observations?
AL	1967	1980	3	Lower Tallapoosa ; Patsaliga ; South Atlantic-Gulf Region
AZ	1975	2024	10	Bill Williams ; Lake Mead ; Lower Colorado ; Lower Gila ; Lower Gila-Painted Rock Reservoir ; Lower Salt ; Middle Gila ; Upper Gila-San Carlos Reservoir ; Upper San Pedro ; Upper Santa Cruz
AR	1998	1998	1	Lower Arkansas
CA	1984	2025	3	Los Angeles ; Lower Colorado ; Seal Beach
CO	1983	1986	2	Alamosa-Trinchera ; San Luis
EL	1961	2025	32	Alafia ; Apalachicola ; Big Cypress Swamp ; Caloosahatchee ; Cape Canaveral ; Charlotte Harbor ; Crystal-Pithlachascotee ; Daytona-St. Augustine ; Everglades ; Florida Southeast Coast ; Hillsborough ; Kissimmee ; Kissimmee ; Lake Okeechobee ; Little Manatee ; Lower St. Johns ; Manatee ; Myakka ; Northern Okeechobee Inflow ; Oklawaha ; Peace ; Peace-Tampa Bay ; Sarasota Bay ; South Atlantic-Gulf Region ; Southern Florida ; St. Johns ; Tampa Bay ; Tampa Bay ; Upper St. Johns ; Vero Beach ; Withlacoochee
GA	1980	1989	3	Altamaha ; Ogeechee Coastal ; South Atlantic-Gulf Region
HI	2018	2018	1	Hawaii
JD	1985	1985	1	Upper Snake-Rock
KS	1967	1990	2	Rattlesnake ; South Fork Ninnescah
LA	2019	2019	1	Amite
NV	1991	2024	4	Lake Mead ; Las Vegas Wash ; Lower Virgin ; Muddy
NC	1965	2018	5	Lower Dan ; Neuse ; Northeast Cape Fear ; Roanoke ; Upper French Broad
OH	2017	2017	1	Cedar-Portage
OK	1977	1999	7	Arkansas-White-Red Region ; Black Bear-Red Rock ; Little ; Lower Canadian-Walnut ; Lower North Canadian ; Polecat-Snake ; Upper Washita
PA	1984	1984	1	Lower Susquehanna
PR	1983	2024	5	Cibuco-Guajataca ; Culebrinas-Guanajibo ; Eastern Puerto Rico ; Puerto Rico ; Southern Puerto Rico
SC	1997	2023	7	Carolina Coastal-Sampit ; Cooper ; Edisto-South Carolina Coastal ; Little Pee Dee ; South Atlantic-Gulf ; South Carolina Coastal ; Waccamaw
TX	1960	2025	48	Amistad Reservoir ; Aransas Bay ; Atascosa ; Austin-Oyster ; Austin-Travis Lakes ; Big Bend ; Bosque ; Brady ; Buchanan-Lyndon B. Johnson Lakes ; Buffalo-San Jacinto ; Cibolo ; Colorado Headwaters ; Elm Fork-Trinity ; Elm-Sycamore ; Guadalupe ; International Falcon Reservoir ; Lake Texoma ; Llano ; Los Olmos ; Lower Brazos-Little Brazos ; Lower Colorado-Cummins ; Lower Devils ; Lower Nueces ; Lower Pecos ; Lower Rio Grande ; Lower Sulphur ; Lower Trinity ; Lower Trinity-Tehuacana ; Lower West Fork Trinity ; Medina ; Middle Brazos-Lake Whitney ; Middle Guadalupe ; Middle Sabine ; Navasota ; Nueces Headwaters ; Rio Grande ; San Ambrosia-Santa Isabel ; San Antonio ; San Marcos ; South Concho ; South Laguna Madre ; Upper Clear Fork Brazos ; Upper Guadalupe ; Upper San Antonio ; Upper Trinity ; West Fork San Jacinto ; West Galveston Bay ; White Oak Bayou

Table last updated 12/8/2025

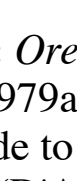
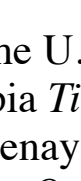
† Populations may not be currently present.

Means of Introduction: This species has been introduced through a combination of means, including stocking and experimental work by states and private companies (e.g., the electric power industry), and release by individuals seeking to use the species as a sport fish, as forage for warmwater predatory fish, as a food source, and as a means of aquatic plant control. Introductions and spread have resulted by way of escapes or releases from aquaculture facilities and experimental control areas, and from various other holding sites (e.g., zoological parks); through aquarium and bait bucket releases; and by intentional transport by anglers and private individuals (Courtney and Hensley 1979a; Lee et al. 1980 et seq.; Courtney et al. 1984, 1986; Muoneke 1988; Courtney and Williams 1992). The exact reasons for and sources of some introductions are uncertain (e.g., Texas) (Hubbs et al. 1978; Courtney and Hensley 1979a). Apparently, power companies became interested in using so-called "tropical fishes" for food or sport in heated effluent ponds used to cool effluents from both fossil fuel fired and nuclear generating plants, where temperatures often became too high to support populations of native fishes (Courtney and Hensley 1979a). Blue tilapia and redbelly tilapia were inadvertently introduced into Hyco Reservoir in North Carolina in 1984 after a small number of fish escaped from a holding cage located in the heated discharge area during an on-site agricultural study (Crutchfield 1995).

Status: Established or possibly established in ten states. Established in parts of Arizona, California, Florida, Nevada, North Carolina, and Texas. Possibly established in Colorado, Idaho, Oklahoma, and Pennsylvania. Reported from Alabama, Georgia, and Kansas. For more than a decade it has been considered the most widespread foreign fish in Florida (Hale et al. 1995).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Other
		

The blue tilapia is considered a competitor with native species for spawning areas, food, and space (Buntz and Manooch 1969; Noble and Germany 1986; Muoneke 1988; Zale and Gregory 1990). Courtney and Robins (1973) reported that certain streams where this species is abundant have lost most vegetation and nearly all native fishes. It has invaded the Taylor Slough portion of Everglades National Park where it is considered a major management problem for the National Park Service (Courtney 1989; Courtney and Williams 1992). The blue tilapia's local abundance and high densities in certain areas have resulted in marked changes in fish community structure (Muoneke 1988, and citations therein). A dramatic reduction in native fishes in the Warm Springs area of Nevada coincided with invasion of this species (Scoppettone et al. 1998, 2005).

Blue tilapia have also been implicated as the cause for unionid mussel declines in two Texas water bodies, Tradinghouse Creek and Fairfield reservoirs (Howells 1995).

Remarks: *Oreochromis aureus* has been used widely in aquaculture and is able to reproduce in brackish waters. The origin of the U.S. stocks of *O. aureus*, was Israel (Courtney and Hensley 1979a). Voucher specimens taken from the lower Colorado river system, Arizona, in 1980 were initially reported as mango tilapia (*Sarotherodon galilaea*; but these were later determined by D. Thys van den Audenaerde to be *O. aureus*). Some lower Colorado River populations in California and Arizona may be hybrids with *O. niloticus* (Courtney et al. 1984, 1986). Although all species from the genus *Oreochromis* readily hybridize (D'Amato et al. 2007), electrophoretic studies on tilapia sampled from 12 Texas reservoirs indicated that most populations were *O. aureus* without indicating genetic introgression with other tilapia species (Howells 1991b). There is a 1971 record of Alabama fish overwintering in outdoor ponds at Auburn University (Courtney and Hensley 1979a, Courtney et al. 1986); however, tilapia introduced into that state typically begin to die the each fall when water temperatures reach about 10°C (Smith-Vaniz 1968). This species was stocked in aquaculture ponds in Iowa to test growth potential, although it reproduced there, it did not overwinter (Pelgren and Carlander 1971; Courtney and Hensley 1979a). In the southwestern United States, the Central Arizona Project canal system is proving to be a major dispersal route for blue tilapia (Courtney, personal communication).

Collection and reported localities were mapped for all or part of the United States, mainly Florida, by Courtney et al. (1974), Foote (1977), Courtney and Hensley (1979a), Lee et al. (1980 et seq.), Loftus and Kushlan (1987), and Menhinick (1991). Hale et al. (1995) reviewed the history of blue tilapia in Florida.

Voucher specimens: Alabama (AUM 1995), Florida (UF 23163, 40306, many others), Kansas (KU 13026, 22917), North Carolina (UNCC uncatalogued), Pennsylvania (PSU 2031).

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Other Resources:
[Global Invasive Species Database Factsheet](#)
[US Fish and Wildlife Service Ecological Risk Screening Summary for Oreochromis aureus](#)
[FishBase Summary](#)

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